

Symbolic Projection Notes for $B_{c1} \rightarrow B_c^{(*)} \gamma$

Working notes

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1 Scope

These notes collect the compact FeynCalc projection results for the heavy-heavy channels

$$B_{c1}(6743) \rightarrow B_c \gamma, \quad B_{c1}(6743) \rightarrow B_c^* \gamma, \quad B_{c1}(6750) \rightarrow B_c \gamma, \quad B_{c1}(6750) \rightarrow B_c^* \gamma.$$

The calculation is structurally different from the D_s and B_s light-cone sum rules. In the B_c system both valence constituents are heavy, so the leading radiative OPE is a hard-photon heavy-heavy calculation. There is no light spectator line to be replaced by photon distribution amplitudes, and no $\chi_s \langle \bar{s}s \rangle$ type term.

2 Currents and Propagators

For $B_c^+ \sim c\bar{b}$, we use

$$j_5 = \bar{b} i \gamma_5 c, \quad j_\rho^V = \bar{b} \gamma_\rho c,$$

and the two axial basis currents

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad J_\mu^B = \frac{i}{m_b + m_c} \bar{b} \sigma_{\mu\alpha} P^\alpha \gamma_5 c, \quad P = p + q.$$

The physical currents are

$$J_{1\mu} = \sin \theta J_\mu^A + \cos \theta J_\mu^B, \quad J_{2\mu} = \cos \theta J_\mu^A - \sin \theta J_\mu^B.$$

This mixing is used at the invariant-amplitude level. The OPE is first derived for the basis currents because the two Dirac traces are independent; the physical amplitudes are then

$$\hat{\Pi}_1 = \sin \theta \hat{\Pi}_A + \cos \theta \hat{\Pi}_B, \quad \hat{\Pi}_2 = \cos \theta \hat{\Pi}_A - \sin \theta \hat{\Pi}_B.$$

For the pseudoscalar final state,

$$g_i = \frac{m_b + m_c}{m_{B_c}^2 f_{B_c} m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c}^2}{M_2^2}\right) \hat{\Pi}_i^{(P)},$$

while for the vector final state,

$$g_i^* = \frac{1}{m_{B_c^*} f_{B_c^*} m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c^*}^2}{M_2^2}\right) \hat{\Pi}_i^{(V)}.$$

The pilot numerical script applies this rotation. The temporary approximation is instead in the normalization: a common benchmark $f_{B_{c1}} = 0.373$ GeV is used for both physical axial-vector residues, pending a separate f_1, f_2 two-point extraction.

The free heavy-quark propagator in momentum space is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0}, \quad Q = c, b.$$

The hard photon is included by inserting the electromagnetic vertex on either heavy line:

$$S_Q^{(\gamma)}(k, k+q) = S_Q^{(0)}(k+q) ie_Q \gamma_\nu \epsilon^\nu S_Q^{(0)}(k).$$

For the $\bar{b}$ line it is convenient to use the effective antiquark charge $e_{\bar{b}} = +e/3$, while $e_c = +2e/3$. Equivalently, one may keep $e_b = -e/3$ and track the orientation of the antiquark line. The scripts use $e_{\bar{b}}$ explicitly.

Thus, yes: this is the same heavy-heavy propagator logic as in a B_c -mixing sum-rule calculation. The important difference from the strange-sector radiative calculation is that we do not insert a light-quark photon DA. If we later include power corrections, the natural heavy-heavy correction is the background-gluon expansion of the heavy propagators, e.g. gluon-condensate terms, not strange-quark condensate photon DAs.

The submitted B_c -mixing work uses the same local basis,

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad J_\mu^B = \frac{i}{m_b + m_c} \bar{b} \sigma_{\mu\alpha} p^\alpha \gamma_5 c,$$

with $m_b = 4.18$ GeV, $m_c = 1.27$ GeV, and the final working window

$$7 \text{ GeV}^2 \leq M^2 \leq 9 \text{ GeV}^2, \quad 53 \text{ GeV}^2 \leq s_0 \leq 55 \text{ GeV}^2.$$

Those are useful starting values for the heavy-heavy OPE in the radiative problem. They are not yet automatically the final radiative-transition windows; the final windows must be fixed by the three-point sum-rule stability, pole dominance, and convergence tests.

3 Step-by-Step Sum-Rule Construction

This section is written as a checklist for reproducing the calculation.

1. Start from basis currents. The physical currents J_1, J_2 are mixed combinations of J_A, J_B . We derive the OPE for J_A and J_B separately only because the Dirac traces are simpler. The physical amplitudes are obtained afterward through the $\sin \theta, \cos \theta$ rotation.

2. Write the correlators. For $B_c \gamma$,

$$\Pi_\mu^{(5,K)}(p, q) = i \int d^4x e^{ipx} \langle \gamma(q, \epsilon) | T \{ \bar{b}(x) i \gamma_5 c(x) J_{K\mu}^\dagger(0) \} | 0 \rangle, \quad K = A, B.$$

For $B_c^* \gamma$,

$$\Pi_{\mu\nu}^{(V,K)}(p, q) = i \int d^4x e^{ipx} \langle \gamma(q, \epsilon) | T \{ \bar{b}(x) \gamma_\nu c(x) J_{K\mu}^\dagger(0) \} | 0 \rangle.$$

3. Contract the heavy fields. There are two hard-photon diagrams: photon emission from the c line and photon emission from the $\bar{b}$ line. The free heavy propagator is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0},$$

and the photon insertion is

$$S_Q^{(\gamma)}(k, k+q) = S_Q^{(0)}(k+q) ie_Q \not{\epsilon} S_Q^{(0)}(k).$$

The charge convention in the scripts is $e_c = +2/3$ and $e_{\bar{b}} = +1/3$. No photon DA or light-quark condensate term appears in this heavy-heavy leading baseline.

4. Evaluate and project the traces. For $B_c\gamma$ project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}, \quad \mathcal{P}^{\mu\nu} = \frac{S^{\mu\nu}}{2(p \cdot q)^2}, \quad S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

For $B_c^*\gamma$ the current diagnostic tensor is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho pq} - (p \cdot q)\epsilon_{\mu\nu\rho p}, \quad V_{\mu\nu\rho}V^{\mu\nu\rho} = -4p^2(p \cdot q)^2.$$

The vector-channel result still needs matching to the physical $1^+ \rightarrow 1^- \gamma$ amplitude basis.

5. Reduce to denominator structures. For photon emission from the charm line use

$$D_1 = (k + q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k - p)^2 - m_b^2.$$

For photon emission from the anti-bottom line use

$$D_1 = k^2 - m_c^2, \quad D_2 = (k - p)^2 - m_b^2, \quad D_3 = (k - p + q)^2 - m_b^2.$$

The triangle pieces contain the full three-denominator product. Terms where a numerator cancels one denominator become two-denominator/contact terms and must be included or shown to vanish.

6. Borel match and rotate. The true variables are p^2 and $P^2 = (p + q)^2$, with

$$p \cdot q = \frac{P^2 - p^2}{2}.$$

The pilot numerical calculation uses the diagonal choice $M_1^2 = M_2^2 = 2M^2$, leading to the practical exponential

$$\exp\left[\frac{m_i^2 + m_f^2}{2M^2}\right].$$

After obtaining $\hat{\Pi}_A, \hat{\Pi}_B$, the physical combinations are

$$\hat{\Pi}_{6743} = \sin\theta \hat{\Pi}_A + \cos\theta \hat{\Pi}_B, \quad \hat{\Pi}_{6750} = \cos\theta \hat{\Pi}_A - \sin\theta \hat{\Pi}_B.$$

For the $B_c\gamma$ widths we use

$$\Gamma(1^+ \rightarrow 0^- \gamma) = \frac{\alpha_{\text{em}}}{3} g^2 E_\gamma^3, \quad E_\gamma = \frac{m_i^2 - m_f^2}{2m_i}.$$

4 $B_{c1} \rightarrow B_c\gamma$: E1 Projection

For the pseudoscalar final state we project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}, \quad \mathcal{P}_{\mu\nu} = \frac{S_{\mu\nu}}{2(p \cdot q)^2}.$$

FeynCalc gives the normalization check

$$S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

We denote

$$p^2 = p_2, \quad p \cdot q = pq.$$

After rewriting loop scalar products in terms of inverse denominators and setting those denominators to zero, the projected triangle-core numerators are

$$C_A^{(c)} = -4i m_c + \frac{4i m_b m_c^2}{pq} - \frac{4i m_c^3}{pq},$$

$$C_A^{(\bar{b})} = -4i m_b + \frac{4i m_b^3}{pq} - \frac{4i m_b^2 m_c}{pq},$$

so that

$$C_A = e_c C_A^{(c)} + e_{\bar{b}} C_A^{(\bar{b})}.$$

For the tensor current,

$$C_B^{(c)} = \frac{4i m_b m_c - 8i m_c^2}{m_b + m_c} - \frac{4i m_c^2 p_2}{(m_b + m_c)pq},$$

$$C_B^{(\bar{b})} = -\frac{4i m_b m_c}{m_b + m_c} - \frac{4i m_b^2 p_2}{(m_b + m_c)pq},$$

and

$$C_B = e_c C_B^{(c)} + e_{\bar{b}} C_B^{(\bar{b})}.$$

The denominator-cancellation terms are stored in the corresponding output file. They should not be discarded automatically; they correspond to reduced two-denominator or contact contributions and must be treated in the dispersion/Borel analysis.

5 $B_{c1} \rightarrow B_c^* \gamma$: Levi-Civita Projection

For the vector final state a useful gauge-invariant structure is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho pq} - (p \cdot q) \epsilon_{\mu\nu\rho p}.$$

It is transverse in the photon index for $q^2 = 0$. The FeynCalc checks are

$$V_{\mu\nu\rho} V^{\mu\nu\rho} = -4p_2(pq)^2, \quad V_{\mu\nu\rho} \mathcal{P}^{\mu\nu\rho} = 1.$$

The projected J_A triangle cores are

$$C_{A,V}^{(c)} = -\frac{4i m_b m_c}{p_2} + \frac{2i m_c^2}{p_2} + \frac{4i m_c^2}{pq},$$

$$C_{A,V}^{(\bar{b})} = -\frac{2i m_b^2}{p_2} + \frac{4i m_b m_c}{p_2} + \frac{4i m_b^2}{pq}.$$

The tensor-current vector cores are

$$C_{B,V}^{(c)} = \frac{2i m_c}{m_b + m_c} + \frac{2i m_b^2 m_c - 4i m_b m_c^2 + 2i m_c^3}{(m_b + m_c)p_2}$$

$$- \frac{4i m_b m_c^2 - 4i m_c^3}{(m_b + m_c)pq} + \frac{2i m_c pq}{(m_b + m_c)p_2},$$

$$C_{B,V}^{(\bar{b})} = \frac{2i m_b}{m_b + m_c} + \frac{-6i m_b^3 + 4i m_b^2 m_c + 2i m_b m_c^2}{(m_b + m_c)p_2}$$

$$- \frac{4i m_b^3 - 4i m_b^2 m_c}{(m_b + m_c)pq} + \frac{2i m_b pq}{(m_b + m_c)p_2}.$$

Again,

$$C_{A,V} = e_c C_{A,V}^{(c)} + e_{\bar{b}} C_{A,V}^{(\bar{b})}, \quad C_{B,V} = e_c C_{B,V}^{(c)} + e_{\bar{b}} C_{B,V}^{(\bar{b})}.$$

6 First Numerical Hard-Photon Pass

The script `Bc_gamma/scripts/bc_radiative_pilot_sumrule.py` uses the compact triangle cores above to produce a first hard-photon numerical estimate. For this pass we use the same central heavy-quark inputs and Borel window as the submitted B_c -mixing analysis,

$$m_b = 4.18 \text{ GeV}, \quad m_c = 1.27 \text{ GeV}, \quad \theta = 43.3^\circ,$$

more precisely $\theta_{\text{mean}} = 43.295^\circ$ with $\sigma_\theta = 0.151^\circ$ and 16–84% interval $[43.147^\circ, 43.455^\circ]$ from the B_c -mixing Monte Carlo. The pilot table uses the rounded central value 43.3° ; the θ uncertainty is not yet included in the quoted radiative intervals.

$$M^2 = \{7, 8, 9\} \text{ GeV}^2, \quad s_0 = \{53, 54, 55\} \text{ GeV}^2.$$

The decay-constant inputs are

$$f_{B_c} = 0.371 \text{ GeV}, \quad f_{B_c^*} = 0.384 \text{ GeV}, \quad f_{B_{c1}} = 0.373 \text{ GeV}.$$

The first two are standard external benchmark inputs. The last is a common axial-vector benchmark used only for this first pass; separate physical f_1 and f_2 residues have not yet been extracted from a two-point mixed-current B_c sum rule in this folder.

Table 1: Pilot hard-photon results in the B_c sector. The intervals show the envelope over $M^2 = \{7, 8, 9\} \text{ GeV}^2$ and $s_0 = \{53, 54, 55\} \text{ GeV}^2$.

Channel	Status	Coupling/projection coefficient	Γ (keV)
$B_{c1}(6743) \rightarrow B_c \gamma$	hard-QCDSR baseline	$+0.0219 [0.0209, 0.0232] \text{ GeV}^{-1}$	$0.108 [0.098, 0.121]$
$B_{c1}(6750) \rightarrow B_c \gamma$	hard-QCDSR baseline	$-0.354 [-0.387, -0.331] \text{ GeV}^{-1}$	$29.6 [25.9, 35.3]$
$B_{c1}(6743) \rightarrow B_c^* \gamma$	diagonal vector pilot	$-0.0620 [-0.0669, -0.0583]$	$46.3 [41.0, 53.8]$
$B_{c1}(6750) \rightarrow B_c^* \gamma$	diagonal vector pilot	$+0.362 [0.335, 0.401]$	$1.67 [1.43, 2.04] \times 10^3$

The $B_c \gamma$ entries are the controlled hard-photon baseline from the present calculation. The $B_c^* \gamma$ entries should be read as diagnostic numbers only. They are obtained from the gauge-invariant $V_{\mu\nu\rho}$ projection and a diagonal single-variable reduction. Before using them as final paper results one must match this coefficient to the physical $1^+ \rightarrow 1^- \gamma$ tensor basis and carry out the full double-Borel analysis, including denominator-cancellation or contact terms.

7 Comparison with Experiment and Other Theory

LHCb has observed a wide peaking structure in the $B_c^+ \gamma$ mass spectrum with significance above seven standard deviations. A minimal two-peak description gives [1, 2]

$$M_1 = 6704.8 \pm 5.5 \pm 2.8 \pm 0.3 \text{ MeV}, \quad M_2 = 6752.4 \pm 9.5 \pm 3.1 \pm 0.3 \text{ MeV},$$

where the errors are statistical, systematic, and due to the B_c^+ mass. No individual radiative partial width is measured. The natural widths are expected to be only a few hundred keV and are neglected in the experimental fit because they are small compared with the photon-energy resolution.

The lower visible peak should not be identified one-to-one with the input 6743 MeV state without care. In the measured $B_c \gamma$ spectrum, decays through $B_c^* \gamma$, followed by an unreconstructed soft $B_c^* \rightarrow B_c \gamma$, can shift visible peaks by the unknown hyperfine splitting.

For comparison with other theory we should not compress different papers into a single range. Table 3 lists each quoted calculation separately. The entries are the values collected in Bondar–Milstein [4]; the original model papers are cited in the first column.

Thus the $B_c \gamma$ baseline is already useful for comparison, while the $B_c^* \gamma$ channel must still be completed with the physical tensor-basis normalization and the full double-Borel/contact-term treatment.

Table 2: Qualitative comparison with the current experimental information from LHCb [1, 2].

Quantity	Experiment	Present input/result
Observed $B_c(1P)^+$ -like signal	yes, $> 7\sigma$ in $B_c^+\gamma$	channels studied explicitly
Lower visible peak	$6704.8 \pm 5.5 \pm 2.8 \pm 0.3$ MeV	input state 6743 MeV
Higher visible peak	$6752.4 \pm 9.5 \pm 3.1 \pm 0.3$ MeV	input state 6750 MeV
Partial widths	not measured	$B_c\gamma$: 0.108, 29.6 keV

Table 3: Comparison with representative predictions for $B_{c1} \rightarrow B_c^{(*)}\gamma$. Widths are in keV.

Publication	$6743 \rightarrow B_c\gamma$	$6743 \rightarrow B_c^*\gamma$	$6750 \rightarrow B_c\gamma$	$6750 \rightarrow B_c^*\gamma$
Godfrey [5]	13	60	80	11
Ebert–Faustov–Galkin [6]	18.4	78.9	132	13.6
Fulcher [7]	32.4	75.6	127.8	26.2
Gershtein et al. [8]	11.6	77.8	131.1	8.1
T. Y. Li et al. [9]	16.6	49	66.6	10.5
Q. Li et al. [10]	35	70	74	40
X. J. Li et al. [11]	30.1	47.8	64	25.6
Eichten–Quigg [12]	9.9	62.5	92.3	7.5
Bondar–Milstein [4]	22	75	47	2
This work, pilot	0.108[0.098, 0.121]	46.3[41.0, 53.8]	29.6[25.9, 35.3]	$1.67[1.43, 2.04] \times 10^3$

8 Perturbative and Condensate Content

The numerical table above contains only the leading hard-photon perturbative baseline. In practical terms this means:

- the photon is emitted from the charm or anti-bottom heavy-quark line;
- both heavy propagators are free propagators in the numerical pilot;
- the retained spectral density is the perturbative triangle hard contribution generated by the projected Dirac traces;
- no photon DA, $\chi\langle\bar{q}q\rangle$, or light-quark condensate contribution appears, because there is no light valence quark in B_c ;
- gluon-condensate corrections from the background-gluon expansion of the heavy propagators are not yet included;
- reduced two-denominator/contact terms from denominator cancellation still require a full double-Borel audit, especially in the vector $B_c^*\gamma$ channel.

Thus the present result is best described as a leading perturbative hard-QCDSR pilot result. A final publication-level B_c analysis should add, or explicitly bound, the heavy-heavy gluon-condensate corrections and the contact terms.

9 Reproduction Checklist for the Present Pilot

The current numbers can be reproduced in the following order.

1. Run `Bc_gamma/scripts/step1_hard_photon_numerators.wl`. This computes the raw FeynCalc traces for photon emission from the charm line and from the anti-bottom line.
2. Run `Bc_gamma/scripts/step2_ps_e1_triangle_reduction.wl`. This projects the $B_c\gamma$ traces onto $S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}$ and checks $S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1$.

3. Run `Bc_gamma/scripts/step2_vec_epsilon_A_reduction.wl` and `Bc_gamma/scripts/step2_vec_epsilon_B_reduction.wl`. These produce the diagnostic $B_c^*\gamma$ cores and check the normalization of $V_{\mu\nu\rho}$.
4. Insert the compact projected cores into `Bc_gamma/scripts/bc_radiative_pilot_sumrule.py`. The script uses the diagonal calibration $M_1^2 = M_2^2 = 2M^2$, with $M^2 = \{7, 8, 9\} \text{ GeV}^2$ and $s_0 = \{53, 54, 55\} \text{ GeV}^2$.
5. Use $m_b = 4.18 \text{ GeV}$, $m_c = 1.27 \text{ GeV}$, and the central mixing angle $\theta = 43.3^\circ$. The current pilot uses $f_{B_c} = 0.371 \text{ GeV}$, $f_{B_c^*} = 0.384 \text{ GeV}$, and a common axial-vector benchmark $f_{B_{c1}} = 0.373 \text{ GeV}$.
6. Rotate the basis amplitudes with $\hat{\Pi}_{6743} = \sin\theta \hat{\Pi}_A + \cos\theta \hat{\Pi}_B$ and $\hat{\Pi}_{6750} = \cos\theta \hat{\Pi}_A - \sin\theta \hat{\Pi}_B$.
7. Convert the pseudoscalar final-state coupling to the width using

$$\Gamma(1^+ \rightarrow 0^-\gamma) = \frac{\alpha_{\text{em}}}{3} g^2 \left(\frac{m_i^2 - m_{B_c}^2}{2m_i} \right)^3.$$

At this stage the $B_c\gamma$ entries are the controlled hard-photon baseline. The $B_c^*\gamma$ entries are useful diagnostics, but they should not be treated as final publication numbers until the physical $1^+ \rightarrow 1^-\gamma$ tensor basis, the full double-Borel treatment, and the reduced-denominator/contact terms have been completed.

10 Next Step

The double-dispersion variables satisfy

$$P^2 = p^2 + 2p \cdot q, \quad pq = \frac{P^2 - p^2}{2}.$$

For $B_c\gamma$, the structure is close to the strange-sector E1 case, but with two heavy masses. For $B_c^*\gamma$, the additional $1/p_2$ and pq/p_2 structures mean that the double dispersion should be written explicitly before numerical Borel analysis. A diagonal single-variable replacement can be useful as a calibration check, but it should not be the final derivation for the vector final state.

Double-dispersion denominator

For the photon emitted from the charm line, the three denominators are

$$D_1 = (k + q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k - p)^2 - m_b^2.$$

With Feynman parameters x, y, z , $x + y + z = 1$, the combined denominator is

$$xD_1 + yD_2 + zD_3 = \ell^2 - \Delta_c,$$

where

$$\Delta_c = (1 - z)m_c^2 + zm_b^2 - zyp^2 - xzP^2.$$

For photon emission from the $\bar{b}$ line, using the momentum routing in the FeynCalc scripts,

$$D_1 = k^2 - m_c^2, \quad D_2 = (k - p)^2 - m_b^2, \quad D_3 = (k - p + q)^2 - m_b^2,$$

and the analogous reduction gives, with $x + y + z = 1$,

$$\Delta_{\bar{b}} = xm_c^2 + (1 - x)m_b^2 - x(1 - x)p^2 + xz(P^2 - p^2).$$

Equivalently one may relabel the Feynman parameters after choosing a different loop momentum. These forms make clear why the true radiative problem is a double-dispersion problem in p^2 and P^2 , not simply the two-point mixing correlator with an extra charge factor.

References

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